

Blockchain

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Abstract— Blockchain technology has become a highly popular industry that has helped a lot of companies advance. Bitcoin is one of the most widely known instances of blockchain being integral in the product. However, there are many other places that blockchain technology can greatly improve. This paper proposes that blockchain technology be implemented in government identification systems for people within the country so that there is a centralized public ledger of all people within the United States for the government to be able to use.

1 PROBLEM SPACE

1.1 Problem and Context

Currently the United States' government does not have an all encompassing database that has entries on every person within the country. The closest databases that are available would be the Social Security Number database and the Internal Revenue Service database, and both of these systems are severely lacking in accuracy of the number of people living in the states. I have personal experience working within the government, and although I am not able to state specific databases, the branch of the government I worked for had access to more than 20 different databases with all different purposes. In addition, there is no easy way of finding or cross referencing individuals within each database since each database was set up differently and collected different information on individuals. This led to many issues of inconsistencies when government workers are working on cases for a person's application in medicare, passports, asylum, driver's license, and many more. In addition, because of the widespread of information across multiple databases, the processes for an individual for anything involving the government will be greatly lengthened due to the need for confirmations across multiple databases and departments. If there was a way for there to not have need for all of the checking and confirming between departments and databases then these processes will be able to be expedited the length of time cut down. The lack of a stable and central system that every government worker can use to look up accurate data on a person simply does not currently exist. This leads to my proposal of the implementation of using blockchain technology to create a blockchain that will act as a system that the government can use to link all of these different databases together.

1.2 Proposed Design

This proposal revolves around blockchain technology becoming the shared centralized system in which people can be identified and data shared about sad people. The Figure below can accurately depict how the government is currently passing information between different groups whenever a need arises. However, with the use of blockchain there would no longer be a need to pass information since every group would have access to the same information if needed.

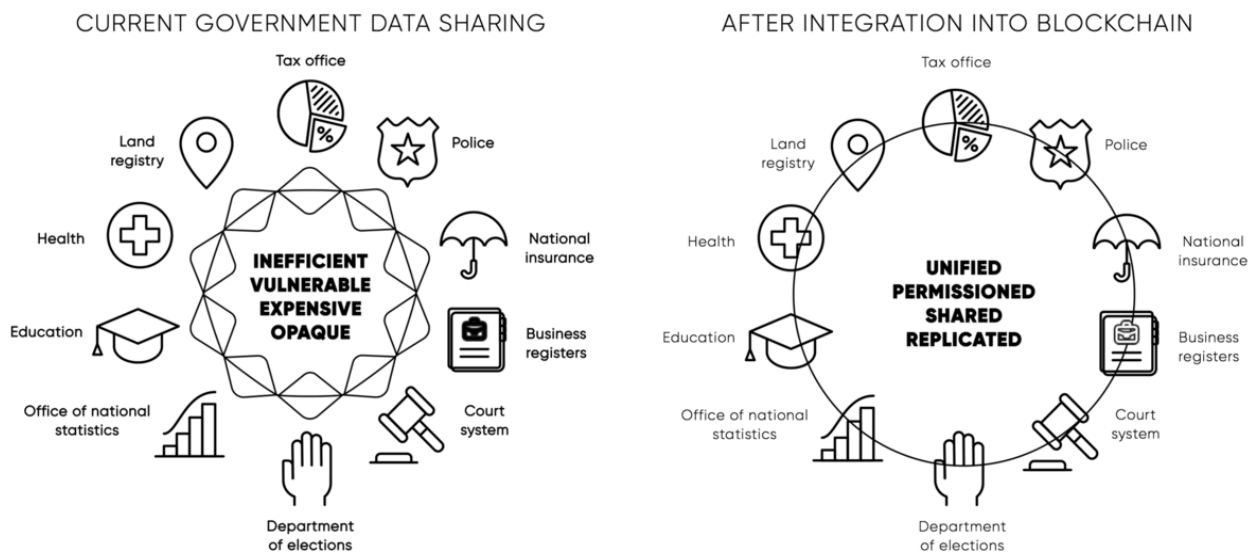


Figure 1—Blockchain influence on Government. [Source](#)

This theory is that a person would take on representation of a bitcoin in current blockchain technology. The person would be able to inherit all the qualities that a bitcoin currently have and also add onto information within the bitcoin through transactions that a bitcoin would normally have. In addition, this would not affect current databases as well since the public key and private key that is used can be substituted for an individual's SSN or ITIN. With this, the blockchain can already be implemented with the user's SSN or ITIN to connect both the IRS and the Social Security Administration's individual databases together. The government can then use this to have the other government departments connect to the user through transactions, which will ultimately create a public ledger of every single person and each department that he or she is involved with, so that everyone can access.

1.3 Comparison

This redesign can be seen as a new implementation of the public's digital identity that can hopefully take over the use of so many multitudes of government databases. This integration of blockchain should be able to minimize inaccuracies found across having multiple databases and

maximize the efficiency of government processes since there is no longer any need for department information sharing.

2 NEED FINDING EXERCISE

2.1 Need Finding

For this needfinding, I interviewed three individuals who all worked for the government involving my proposal. Since all three individuals work within the government, they are able to give me an alternative view on how they have interacted with all the different types of databases they use. In addition, all three individuals work for different branches of the government which will give me a wider array of responses to support my proposal. For each person, I asked them their thoughts on the current process of how their current work is involved in databases and how they are affected by the databases. Then I described my proposal to them and asked them for their input. The first individual worked closely with individuals applying for immigration and citizenship, which resulted in them constantly working with several databases involving many individuals. They have explicitly stated that the excessive number of databases is a great hindrance to their work and the proposal seems like an excellent plan. The second individual worked within the technology departments of the government. Their response was very similar to the first person since although they did not work with the people themselves, they agreed with the process and believe that this proposal is an improvement to the multitudes of databases. The third person worked within the FDA and did not have any involvement with any databases involving people. This person's response was completely neutral since their branch of the government did not involve working with the people themselves. This exercise shows that within the individuals interviewed, the two individuals who had the most exposure to government databases supported the proposal for the blockchain technology implementation.

2.2 Baseline Design

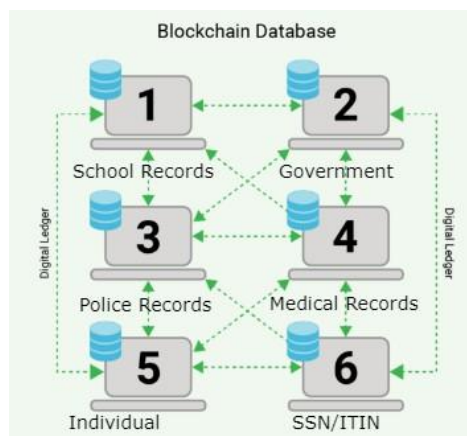


Figure 2—Public ledger of the people

Public key of user - random generated key that's public

Private key of user - the user's privately created key from user SSN/ITIN

Individual Information sent during transaction - All user information, applications, all user associated relevant past information

2.3 Architecture

The process in which the transaction creates the public ledger, in this example will be an individual applying for medicare:

1. User creates a transaction
2. Transaction includes: user's public key (address) and the recipient's public key (address), the individual's information and application (in this situation can be viewed as bitcoin)
3. Transaction must be signed with digital signature using user's private key
4. Broadcasting and verification (this process follows the same steps as in bitcoin)
5. New block is added to the blockchain and the transaction is complete

The next steps would be for the government department handling the medicare request to send a transaction back with acceptance or rejection along with all the relevant individual's information back and everything within the process will be recorded within the public ledger of the block chain.

2.4 Alternative Design

An alternative design to this could be the change how the individual is viewed within the blockchain itself. A clear flaw within this current design is the exaggeration that the individual themselves is a singular bitcoin in the example. However, this causes sewer backup in a person who needs to make multiple transactions at the same time since the user will not be able to use his or her bitcoin again until the transaction is sent back. An alternative design is to have the users an entity that sends and receives multiple transactions unencumbered since the transaction is severely limited in information and the ledger holding all the information instead. This alternative design can be seen in the next section.

3 DESIGN REFINEMENTS

3.1 Refinements

The baseline design is further improved in this different refined design. It was clear that the baseline design was functional but can cause issues of efficiency when it comes to the number of transactions possible from an individual. The refinements took the original process and made changes to the transaction to give more responsibility back to the people. The transactions would

no longer hold all the information of the user, but instead only hold important user information and the relevant documents instead, so to get all the information on the user the government would need to look through the blockchain to receive all of the information on the user. This will allow the user to create multiple transactions all at once without having to wait for individual departments to send transactions back.

3.2 Functional Components

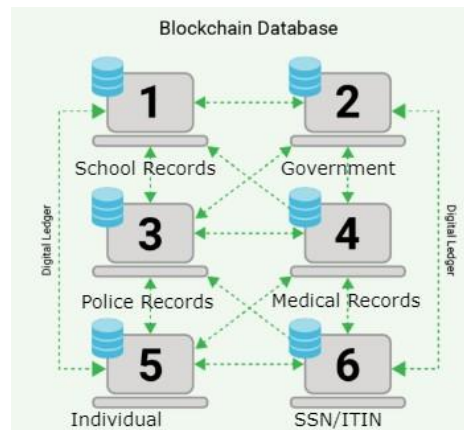


Figure 2—Public ledger of the people

Public key of user - random generated key that's public

Private key of user - the user's privately created key from user SSN/ITIN

Individual Information sent during transaction - Only relevant user information, applications

3.3 Architecture

The process in which the transaction creates the public ledger, in this example will be an individual applying for medicare:

1. User creates a transaction
2. Transaction includes: user's public key (address) and the recipient's public key (address), the individual's application (in this situation can be viewed as bitcoin)
3. Transaction must be signed with digital signature using user's private key
4. Broadcasting and verification (this process follows the same steps as in bitcoin)
5. New block is added to the blockchain and the transaction is complete

The next steps would be for government department receiving the transaction. The government must then take the information about the user from the transaction and look through the public ledger to compile all the user's information. Then the government would be handling the medicare request to send a transaction back with acceptance or rejection along with all the

relevant individual's information back and everything within the process will be recorded within the public ledger of the block chain.

3.4 Improvements from Baseline

Although the baseline design and the refined design are very similar in the process, the main difference is in the transaction and use of the public ledger. The transaction will no longer contain all the user's past information and only the relevant information for the specific department. The department would need to look through the blockchain themselves to retrieve all the relevant past information if needed. This puts a lot more emphasis on the public ledger that records all the transactions. In addition, the government themselves would need to take on more responsibility in checking transactions and looking through the ledger themselves. However, this downside would allow the people to be able to make multiple transactions to different departments, which the baseline design does not allow.

3.5 Evaluation metrics

The refined design will be evaluated on both efficiency and functionality through continued use to ensure that the design itself is an improvement to both the original system and the baseline design. The evaluation for functionality will be heavily dependent on whether government workers would be able to continue to complete their assignments without hindrance. Currently, government workers are able to complete their work without issue. However, the goal is to be able to continue to keep functionality working properly after the design implementation. The next evaluation will be based on efficiency. This will depend on the time in which government employees are able to complete his or her work faster than without the blockchain integration. If workers are able to complete their work faster than without the new blockchain technology then the design can be considered a success.

4 ADDITIONAL DESIGN CONSIDERATIONS

4.1 Evaluation of Baseline

The baseline design will be evaluated similarly to the refined design on functionality. Since its the baseline design, the main focus will be on functionality and whether or not the system is actually a viable fix to the current issue that government workers encounter. The evaluation will be based on whether or not a government worker is able to share information with another department on individuals without having to reach out directly. As long as information sharing is able to spread throughout all the departments accurately then the functionality of the design can be ensured. The speed and efficiency of the design is not as important for the baseline design compared to the refined design however the efficiency of the baseline design cannot drop below the original system. The speed will be based on the length of the entire process in which an

individual is able to apply for some type of government process and have the process completed. This will determine the speed of the process which can be compared to the normal processing time of the application. As long as the speed and functionality of the baseline can be kept the same or better than the original system, the baseline design can be considered to be successful.

4.2 Additional Functional Requirements

Additional functional requirements of the design may be the possible inclusion of possible cases in which private information is lost or blockchain information needed to be changed. Blocks within a block chain were designed to be extremely difficult to change. However, there will always be situations where a change can be needed so a system in which blocks can be manipulated in special circumstances will be needed. In addition, a user may lose his or her private key. This situation will also call for some type of recovery of the key to be implemented. Stolen identities will be more difficult however it could still be possible. There needs to be a system in place to check and lock down stolen identities if possible. All of these situations require a system to be created to assist with these instances in further possible redesigns.

5 REFERENCES

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